

Multiple Choice

1. **Answer:** A

Options: A) 108 nucleotides; B) 120 nucleotides; C) 150 nucleotides; D) 144 nucleotides; E) 60 nucleotides

Note: Correct option: A - 108 nucleotides

2. **Answer:** A

Options: A) Cortisol, aldosterone, and adrenal androgens; B) Insulin, glucagon, and somatostatin; C) Cholecystokinin, ghrelin, and leptin; D) Vasopressin, erythropoietin, and renin; E) Epinephrine, norepinephrine, and dopamine

Note: Correct option: A - Cortisol, aldosterone, and adrenal androgens

3. **Answer:** E

Options: A) Epilepsy is the result of a chromosomal aneuploidy, similar to Down syndrome.; B) Epilepsy is always inherited in an X-linked recessive pattern.; C) Epilepsy is a dominant trait.; D) Epilepsy is caused by an autosomal dominant trait.; E) Epilepsy is due to an autosomal recessive trait.

Note: Correct option: E - Epilepsy is due to an autosomal recessive trait.

4. **Answer:** E

Options: A) Lysosome; B) Chloroplast; C) Ribosome; D) Golgi apparatus; E) Plasma membrane

Note: Correct option: E - Plasma membrane

5. **Answer:** D

Options: A) Centrioles; B) Chloroplasts; C) Lysosomes; D) Cell wall; E) Mitochondria

Note: Correct option: D - Cell wall

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'They have flowers that bloom for several years before producing seeds'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Complete equilibrium in a gene pool is not expected'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** (a) 4 chromatids, (b) 12 chromatids, (c) 6 chromatids. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Temperature, precipitation, soil chemistry, and the types of decomposers present. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key